

4.3 Linearly Independent Sets; Bases

An indexed set of vectors $\{\vec{v}_1, \dots, \vec{v}_p\}$ is said to be linearly independent if the vector equation

$$c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_p \vec{v}_p = \vec{0} \quad (1)$$

has only the trivial solution, $c_1=0, \dots, c_p=0$.

The set $\{\vec{0}\}$ is thus linearly dependent because

$$1 \cdot \vec{0} = \vec{0}$$

which is not the trivial solution.

Theorem 4

An indexed set $\{\vec{v}_1, \dots, \vec{v}_p\}$ of two or more vectors, with $\vec{v}_1 \neq \vec{0}$, is linearly dependent if and only if some $\vec{v}_j$ ($j > 1$) is a linear combination of the preceding vectors, $\vec{v}_1, \dots, \vec{v}_{j-1}$.

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Example 1

Let $\vec{p}_1(t) = 1$, $\vec{p}_2(t) = t$, and $\vec{p}_3(t) = 4-t$. Then

$\{\vec{p}_1, \vec{p}_2, \vec{p}_3\}$ is linearly dependent in $\mathbb{P}$ because

$$\vec{p}_3 = 4\vec{p}_1 - \vec{p}_2.$$

Example 2

The set $\{\sin t, \cos t\}$ is linearly independent in $C[0,1]$, the space of all continuous functions on $0 \leq t \leq 1$, because $\sin t$ and $\cos t$ are not multiples of one another as vectors in $C[0,1]$. In contrast, $\{\sin t \cos t, \sin 2t\}$ is linearly dependent because of the identity $\sin 2t = 2 \sin t \cos t$, for all t .

Definition

Let H be a subspace of a vector space V . A set of vectors β in V is a basis for H if

- (i) β is a linearly independent set, and
- (ii) the subspace spanned by β coincides with H ; that is,

$$H = \text{Span } \beta$$

Example 3

Let A be an invertible $n \times n$ matrix, $A = [\vec{a}_1 \dots \vec{a}_n]$. Then the columns of A form a basis for $\mathbb{R}^n$ because they are linearly independent and span $\mathbb{R}^n$, by the invertible matrix theorem.

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Example 4

Let $\vec{e}_1, \dots, \vec{e}_n$ be the columns of the $n \times n$ identity matrix I_n . That is

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, \vec{e}_n = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

The set $\{\vec{e}_1, \dots, \vec{e}_n\}$ is called the standard basis for $\mathbb{R}^n$ (Figure 1).

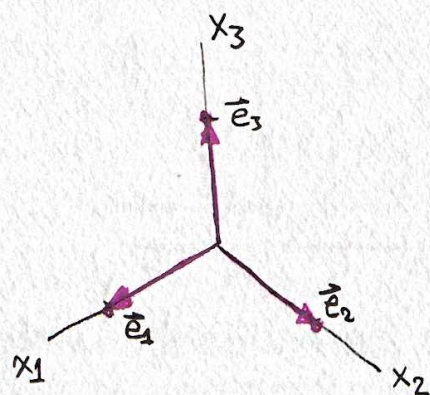


Figure 1 - standard basis for $\mathbb{R}^3$

Example 5

Let $\vec{v}_1 = \begin{bmatrix} 3 \\ 0 \\ -6 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} -4 \\ 1 \\ 7 \end{bmatrix}$, and $\vec{v}_3 = \begin{bmatrix} -2 \\ 1 \\ 5 \end{bmatrix}$. Determine

if $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is a basis for $\mathbb{R}^3$.

Solution

Let $V = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$. Is V linearly independent? Row reduction gives

$$\begin{bmatrix} 3 & -4 & -2 \\ 0 & 1 & 1 \\ -6 & 7 & 5 \end{bmatrix} \begin{array}{l} r_3 \leftarrow r_3 + 2r_1 \\ r_2 \leftarrow r_2 + r_3 \end{array} \sim \begin{bmatrix} 3 & -4 & -2 \\ 0 & 0 & 2 \\ 0 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The set V is linearly independent, thus V is a basis for $\mathbb{R}^3$.

Example 6

Let $S = \{1, t, t^2, \dots, t^n\}$. Verify that S is a basis for $\mathbb{P}_n$. This basis is called the standard basis for $\mathbb{P}_n$.

Solution

Certainly S spans $\mathbb{P}_n$. To show that S is linearly independent, suppose that $c_0, \dots, c_n$ satisfy

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$$c_0 1 + c_1 t + c_2 t^2 + \dots + c_n t^n = \vec{0}(t) \quad (2)$$

This equality means that the polynomial on the left has the same value as the zero polynomial on the right. A fundamental theorem in algebra says that the only polynomial in $\mathbb{P}_n$ with more than n zeros is the zero polynomial. That is, equation (2) holds for all t only if $c_0 = \dots = c_n = 0$. This proves that S is linearly independent and hence is a basis for $\mathbb{P}_n$. See Figure 2.

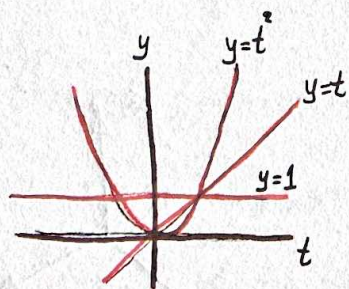


Figure 2 - The standard basis for $\mathbb{P}_2$.

The Spanning Set Theorem

A basis is a spanning set containing no unnecessary vectors; that is, it contains only the vectors necessary to do the spanning.

Example 7

Let $\vec{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$, $\vec{v}_3 = \begin{bmatrix} 6 \\ 16 \\ -5 \end{bmatrix}$, and $H = \text{Span}\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$.

Note that $\vec{v}_3 = 5\vec{v}_1 + 3\vec{v}_2$, and show that $\text{Span}\{\vec{v}_1, \vec{v}_2, \vec{v}_3\} = \text{Span}\{\vec{v}_1, \vec{v}_2\}$. Then find a basis for the subspace H .

Solution

Every vector in $\text{Span}\{\vec{v}_1, \vec{v}_2\}$ belongs to H because

$$c_1 \vec{v}_1 + c_2 \vec{v}_2 = c_1 \vec{v}_1 + c_2 \vec{v}_2 + 0 \vec{v}_3$$

Now let $\vec{x}$ be any vector in H . - say $\vec{x} = c_1 \vec{v}_1 + c_2 \vec{v}_2 + c_3 \vec{v}_3$.

Since $\vec{v}_3 = 5\vec{v}_1 + 3\vec{v}_2$, we may substitute

$$\begin{aligned} \vec{x} &= c_1 \vec{v}_1 + c_2 \vec{v}_2 + c_3 (5\vec{v}_1 + 3\vec{v}_2) \\ &= (c_1 + 5c_3) \vec{v}_1 + (c_2 + 3c_3) \vec{v}_2 \end{aligned}$$

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Thus $\vec{x}$ is in $\text{Span}\{\vec{v}_1, \vec{v}_2\}$, so every vector in H already belongs to $\text{Span}\{\vec{v}_1, \vec{v}_2\}$. Obviously linearly independent, it follows that $\{\vec{v}_1, \vec{v}_2\}$ is a basis for H . See figure 3.

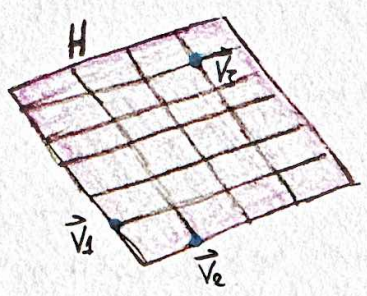


Figure 3 - basis for H .

The next theorem generalizes Example 7.

Theorem 5 - The Spanning Set Theorem

Let $S = \{\vec{v}_1, \dots, \vec{v}_p\}$ be a set in a vector space V , and let $H = \text{Span}\{S\}$.

- If one of the vectors in S - say, $\vec{v}_k$ - is a linear combination of the remaining vectors in S , then the set formed from S by removing $\vec{v}_k$ still spans H .
- If $H \neq \{\vec{0}\}$, some subset of S is a basis for H .

Bases for Nul A, Col A, and Row A

We already went through how to find basis for Nul A, look at Example 3 p.237 in the text book.

In the following two examples we will describe a simple algorithm for finding a basis for the column space.

Example 8

Find a basis for Col B, where

$$B = [\vec{b}_1 \ \vec{b}_2 \ \dots \ \vec{b}_5] = \begin{bmatrix} \textcircled{1} & 4 & 0 & 2 & 0 \\ 0 & 0 & \textcircled{1} & -1 & 0 \\ 0 & 0 & 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution

Each nonpivot column of B is a linear combination of the pivot columns. We identify

$$\vec{b}_2 = 4\vec{b}_1, \vec{b}_4 = 2\vec{b}_1 - \vec{b}_3$$

thus $\vec{b}_1, \vec{b}_3$ and $\vec{b}_5$ are pivot columns. By theorem 5, we have

$$S = \{\vec{b}_1, \vec{b}_3, \vec{b}_5\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\} \text{ is a spanning set for basis of Col B.}$$

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Example 9

It can be shown that the matrix

$$A = [\vec{a}_1 \ \vec{a}_2 \ \dots \ \vec{a}_5] = \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 3 & 12 & 1 & 5 & 5 \\ 2 & 8 & 1 & 3 & 2 \\ 5 & 20 & 2 & 8 & 8 \end{bmatrix} \text{ is row equivalent}$$

to B in Example 8. Find a basis for $\text{Col } A$.

Solution

From Example 8 we learned that

$$\vec{a}_2 = 4\vec{a}_1 \text{ and } \vec{a}_4 = 2\vec{a}_1 - \vec{a}_3,$$

and upon inspection we can verify this.

This means $\{\vec{a}_1, \vec{a}_3, \vec{a}_5\}$ is a basis for $\text{Col } A$.

Warning: Row operations can change the column space of A matrix. This means $\text{Col } B$ may not equal $\text{Col } A$, where B is matrix A row reduced to echelon form.

Theorem 6

The pivot columns of A forms a basis for $\text{Col } A$.

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In contrast, the following theorem establishes that row reduction does not change the row space of a matrix.

Theorem 7

If two matrices A and B are row equivalent, then their row spaces are the same. If B is in echelon form, the nonzero rows of B form a basis for the row space of A as well as for that of B .

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Example 10

Find a basis for the row space of A from Example 9.

Solution

Remember that $A \sim B$, and from Theorem 7 we have that all nonzero rows of B forms a basis for $\text{Row } B$ as well as $\text{Row } A$.

Thus, basis for $\text{Row } A$:

$$\{(1, 4, 0, 2, 0), (0, 0, 1, -1, 0), (0, 0, 0, 0, 1)\}$$

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Two Views of a Basis

When the spanning set theorem is used, the deletion of vectors from a spanning set must stop when the set becomes linearly independent. If an additional vector is deleted, it will not be a linear combination of the remaining vectors, and hence the smaller set will no longer span V .